The four predictors in one page: inputs, assumptions, tests, limits

All numbers are Brier ALC from the online simulation (lower is better); "minus BB" is the test difference to the Beta-Binomial with a 95% bootstrap CI over tasks (paired). Test uses 3 never-trained benchmarks (`algotune`, `cmmlu`, `editbench`); validation uses `ai2d_test`, `bird_sql`, `financebench`. No model here uses subject metadata or a benchmark label as a predictive feature; they serve only to select or index labels (the hierarchical Beta-Binomial, not shown, did use metadata). Inputs are read from the submission code. Items marked *(reasoning)* are my deduction from the equations, not something we tested.

	Beta-Binomial (baseline)	**ANP**	**Pooled Rasch**	**Kernel Rasch**
Labels used	only this subject's labels on this benchmark (counts h, n)	only this subject's labels on this benchmark, as the context (a fixed maximum number of the latest)	every revealed label in the draw: all subjects, all benchmarks	same as pooled Rasch
Subject input	none; the subject record only selects which labels are its own	none enters the network. The subject record (name, provider, dates, harness, ...) is used only to decide which labels belong to the same subject; its contents are never read	a name tag only: the whole subject record (name, provider, dates, harness, ...) is turned into one piece of text; labels with the same text belong to one subject and share one ability θ_s. The contents are never read, so two subjects that differ only in provider are unrelated	same as pooled Rasch
Benchmark input	none; `benchmark_id` only selects the labels	none enters the network; `benchmark_id` only selects same-benchmark labels	`benchmark_id` is a name tag: labels with the same `benchmark_id` share one benchmark rate μ_b	the same name tag for μ_b; it also decides which items share the kernel K (only items of one benchmark are linked)
Item input	none (item text ignored)	BGE-M3 embedding of `item_content`, for the target and every context item	item identity only: a hash of `benchmark_id`, `item_content`, `item_features`; no embedding	the same identity, plus the BGE-M3 embedding of `item_content`
Output	one probability for every item of the task	one probability per item	one probability per (subject, item)	same
Model	$\hat{p} = (h + 0.5)/(n + 1)$	latent attentive neural process: 1 self-attention block over the context, 4-head cross-attention from each target, 32-d latent, width 64, MLP decoder	$\operatorname{logit} p = \mu_b + \theta_s - b_i$; $\mu_b \sim \mathcal{N}(m_0, 2^2)$, $m_0 = \operatorname{logit}(0.512)$; $\theta_s \sim \mathcal{N}(0,1)$; $b_i \sim \mathcal{N}(0,1)$	same, but $\mathbf{b} \sim \mathcal{N}(0, K)$ within each benchmark, $K_{ij} = \tau^2\left[(1-\rho)\delta_{ij} + \rho\, e^{(\cos(x_i, x_j)-1)/T}\right]$; unlabelled item: $\hat{b}_i = K_{iL} K_{LL}^{-1} \hat{\mathbf{b}}_L$
Fitted	nothing	offline on 11 training benchmarks (negative ELBO, random context sizes 0–31), early-stopped on a fixed validation bank, one seed. Variants: + SAM, prefix training	no offline training; 3 prior widths picked on validation; MAP (Newton in the submission, L-BFGS in the simulation), refit from scratch at budgets 0, 1, 3, 7, 15, 31	no offline training; $\tau = 1.4$, $\rho = 0.8$, $T = 0.2$ picked on 60 validation draws (at the edge of the grid); otherwise as pooled
Assumes	one exchangeable accuracy per task, independent labels	test tasks resemble training tasks; items enter only through the embedding; weights frozen at test time	equal item discrimination (Rasch); labels independent given parameters; the same item has the same difficulty for every subject; Gaussian priors; an unlabelled item has difficulty 0	pooled's assumptions, but difficulty is not independent: text-similar items have similar difficulty, decaying with cosine distance; no sharing across benchmarks (block-diagonal)
Tested	single-pair 82 val / 45 test tasks; multi-pair 60 val / 150 test draws	single-pair only: test ALC 0.2725 (ANP), 0.2663 (+SAM), 0.2762 (prefix training) vs 0.2494; also multi-pair for the plain ANP (0.2726)	multi-pair, 150 test draws, random acquisition: 0.2461, **minus BB -0.0035 [-0.0046, -0.0026]**	same draws: 0.2448, **-0.0049 [-0.0060, -0.0038]**; minus pooled -0.0013 [-0.0017, -0.0010]
Competition-shaped (one subject per benchmark, 150 test draws)	reference	not run	-0.0011 with old priors; **-0.0032** with priors re-tuned on 120 validation draws	**-0.0024**; on validation in this shape it ties BB (0.1735 vs 0.1738) and pooled does not beat it
Known limits	answers are not independent: text-similar items correlate $+0.24$ on test; ignores item content	every ANP is significantly worse than BB on test, $+0.0168$ to $+0.0267$; the loss sits at 0–1 labels (ability component miscalibrated $+0.04$ to $+0.06$); item ordering does not transfer (OOD rank correlation about 0.00–0.04); training is 71% one benchmark (`alignment_faking`)	ability is estimated no better than BB (RMSE 0.18 vs 0.13 at one label); with one subject per benchmark μ_b and θ_s are not separately identified, the priors split them *(reasoning)*	needs BGE-M3 inside the submission; much of the effect on `editbench` is near-duplicate items (54% have cosine > 0.97; 0–4% elsewhere); text embeddings recover little cross-benchmark difficulty (probe OOD rank correlation 0.09)

Not tested, for any of these: leave-one-benchmark-out (only 3 test benchmarks, so results can turn on which are held out); per-benchmark loss weighting; more than one ANP seed; acquisition policies other than random for the Rasch models (only one diverse-acquisition run, no effect); the competition's real hidden draw (about 7 benchmarks; unpaired leaderboard: pooled Rasch 0.2022 vs BB 0.1486; kernel Rasch submission 1026512 unscored); single-pair runs of the Rasch models; text-entropy features (computed on question text, not used in any model).